Speaker Notes
AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media

Diploma defence speech notes

Bachelor's diploma defence, 2026

Contents

Slide 1. AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media

Purpose:

Open the defence, name the topic, and immediately set the correct research scope.

What to say:

Good morning. The topic of our diploma work is *AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media*. The work studies how machine learning can support directional prediction for thermoelastic wave behavior in simplified geological media. I would like to emphasize the scope from the beginning: this is a research prototype and a comparative framework. It is not presented as a site-calibrated simulator or as a replacement for numerical methods. The focus is on physical interpretation, model comparison, and estimating directional characteristics under simplified assumptions.

Key points:

- The work connects thermoelastic physics, geological parameters, and neural surrogate models.
- The main target is directional prediction between a source and a probe point.
- The presentation treats the system as a research prototype.

Transition to next slide:

I will start by explaining why this topic is relevant.

Slide 2. Motivation and Research Problem

Purpose:

Open the introduction with the motivation, the research problem, and the correct prototype scope.

What to say:

This slide combines the motivation and the problem statement. Thermoelastic wave propagation is relevant because temperature changes in rocks can influence stress, deformation, and wave behaviour. This matters in geothermal systems, underground engineering, and geophysical interpretation. The research problem is to estimate directional characteristics of thermoelastic wave propagation using an AI-assisted comparative framework under simplified physical assumptions. The image reminds us that real rocks are complex, but in this thesis they are represented through effective material parameters.

Key points:

- Temperature changes in rocks can produce mechanical effects.
- The task is directional source–probe prediction, not full field-scale simulation.
- The work is a research prototype, not an industrial geophysical simulator.

Transition to next slide:

After defining the problem, I will state the aim, tasks, object, and subject of the work.

Slide 3. Aim, Tasks, and Scope

Purpose:

State the aim and compress the thesis tasks into one defensible overview slide.

What to say:

The aim is to develop and justify an AI-guided framework for directional prediction of thermoelastic wave propagation in geological media. The work first defines the physical and geological basis, then formulates a simplified two-dimensional thermoelastic prediction task. After that, it prepares COMSOL-derived synthetic data, compares four surrogate routes, and interprets the results and limitations. The object is thermoelastic wave propagation in geological media. The subject is AI-assisted directional prediction along a selected source–probe path. The temperature-field image illustrates the controlled 2D simulation basis used for comparison.

Key points:

- Aim: AI-guided directional thermoelastic wave prediction.
- Tasks: physical basis, 2D formulation, COMSOL data, model comparison, interpretation.
- Object and subject are stated explicitly on the slide.

Transition to next slide:

Now I will explain the basic physical logic of thermoelastic coupling.

Slide 4. Thermoelastic Coupling Logic

Purpose:

Introduce the physical chain from heating to wave-like mechanical response.

What to say:

This slide shows the basic logic of thermoelastic coupling. A local thermal source changes the temperature field. Temperature change produces thermal strain, and if the expansion is constrained by the surrounding material, stress appears. Stress gradients then create mechanical displacement, and this response propagates through the medium. The chain at the bottom is a physical summary rather than a full equation: T is temperature in kelvin, $\varepsilon_{\mathrm{th}}$ is dimensionless thermal strain, σ is stress in pascals, and u is displacement in metres.

Key points:

- Heating changes the temperature field.
- Thermal expansion can produce stress.
- Stress gradients drive displacement and propagation.

Transition to next slide:

The next group of slides explains the physical and geological parameters used in this chain.

Slide 5. Geological Media and Effective Parameters

Purpose:

Show that geological materials enter the prototype through effective scalar parameters.

What to say:

Here we show a compact example of the geological material parameters used in the study. Density, compressional wave velocity, thermal conductivity, and heat capacity differ between rock types such as granite, basalt, sandstone, and limestone. In the prototype, these are effective scalar parameters. They allow us to compare material behavior in a controlled way. The limestone texture is included only as an illustrative geological material image; it is not used as input data.

Key points:

- Rock types differ in density, stiffness, wave speed, and thermal properties.
- The parameters are effective values for comparison.
- The table supports modelling assumptions rather than field calibration.

Transition to next slide:

The next slide separates the mechanical part of the material catalogue.

Slide 6. Mechanical Rock Parameters

Purpose:

Document the mechanical part of the material catalogue used for comparison.

What to say:

This table separates the mechanical parameters from the larger material catalogue. It lists density, Young's modulus, Poisson's ratio, shear modulus, bulk modulus, and representative compressional and shear wave velocities. I will not read every value. The purpose is to show that the model inputs are physically meaningful and material-dependent, while still being effective midpoint values used under isotropic assumptions.

Key points:

- Mechanical parameters control inertia, stiffness, and elastic wave speeds.
- The values are effective and comparative, not site-calibrated.
- Vertical table boundaries separate values and units for readability.

Transition to next slide:

The next slide gives the thermal and porosity part of the catalogue.

Slide 7. Thermal and Porosity Parameters

Purpose:

Document the thermal and porosity parameters without overloading one table.

What to say:

This table gives thermal expansion, thermal conductivity, heat capacity, volumetric heat capacity, thermoelastic coupling coefficient, and porosity information. These quantities control how a temperature disturbance spreads and how temperature change contributes to thermal strain and stress.

Missing values are shown as dashes rather than filled by unsupported assumptions. This keeps the material representation transparent.

Key points:

- Thermal parameters control heat spreading and thermal strain.
- Porosity is included as an effective descriptive property.
- Missing values are explicit and are not silently invented.

Transition to next slide:
After listing the parameters, I will explain how they control propagation.

Slide 8. Rock Properties Controlling Propagation

Purpose:
Explain the physical role of mechanical and thermal rock properties.

What to say:
Rock properties influence propagation through both mechanical and thermal mechanisms. Density controls inertia: a denser medium resists acceleration more strongly. Elastic stiffness controls how strongly the rock resists deformation. Porosity, cracks, and saturation can reduce effective stiffness and change wave behavior. On the thermal side, conductivity controls heat transfer, heat capacity controls thermal storage, and thermal expansion controls how temperature change becomes strain. In real rocks, bedding, foliation, aligned pores, fractures, and stress state can also create directional effects.

Key points:

- Mechanical properties control inertia and stiffness.
- Thermal properties control heat spreading and thermal strain.
- Geological structure can create directional behavior.

Transition to next slide:
To connect this with wave propagation, I will briefly discuss P-waves and S-waves.

Slide 9. Elastic Waves in Geological Media

Purpose:
Introduce compressional and shear waves using simple physical interpretation.

What to say:
Elastic wave propagation is commonly described using compressional waves and shear waves. Compressional waves have particle motion mainly parallel to the propagation direction, while shear waves have transverse motion and depend on shear rigidity. The formulas define V_P and V_S in metres per second. They depend on bulk modulus K, shear modulus G, and density ρ. In physical terms, larger stiffness tends to increase wave velocity, while larger density increases inertia.

Key points:

- P-waves involve compression and volume change.

- S-waves depend on shear rigidity.
- Wave velocities depend on elastic moduli and density.

Transition to next slide:

Next, I will add the thermal part of the coupled problem.

Slide 10. Heat Conduction and Thermal Expansion

Purpose:

Explain how heat transfer enters the thermoelastic formulation.

What to say:

The thermal part is described by Fourier heat flux and the heat equation. In the notation, q is heat flux in watts per square metre, k is thermal conductivity, T is temperature in kelvin, ρ is density, C_p is heat capacity, and Q is a volumetric heat source. The second relation defines temperature perturbation and thermal strain: θ and ΔT are temperature changes in kelvin, α is thermal expansion, and $\varepsilon_{\mathrm{th}}$ is dimensionless thermal strain. The physical meaning is simple: heat diffuses through the rock, and temperature increase can create strain.

Key points:

- Thermal conductivity controls the heat flux.
- The heat equation describes temperature evolution.
- Thermal expansion links temperature change to strain.

Transition to next slide:

Now I will summarize the key physical parameters in one place.

Slide 11. Key Physical Parameters

Purpose:

Make the parameter roles easy to remember for the committee.

What to say:

This slide summarizes thermal diffusivity. The relation $\alpha_T = k/(\rho C_p)$ combines thermal conductivity, density, and heat capacity. The result has units of square metres per second and describes how quickly a temperature disturbance spreads through the medium. This parameter is useful because it turns several material properties into one interpretable quantity related to thermal spreading.

Key points:

- Thermal diffusivity describes the rate of heat spreading.
- Density and stiffness influence mechanical wave behavior.
- Thermal expansion is the bridge from heat to deformation.

Transition to next slide:

The next slide combines these ideas in the thermoelastic equations.

Slide 12. Thermoelastic Coupling

Purpose:

Present the theoretical backbone without overloading the speech with derivation.

What to say:

This slide gives the thermoelastic extension of Hooke's law. The first relation defines temperature perturbation relative to the reference temperature. The stress relation contains the elastic part, controlled by the Lame parameters, and the thermal part, controlled by the coupling coefficient $\gamma = 3K\alpha$. Stress is measured in pascals, strain is dimensionless, and displacement is measured in metres. The equation of motion then links stress gradients to acceleration and displacement. In simple terms, temperature changes can modify stress, and stress drives motion.

Key points:

- Stress contains both elastic and thermal contributions.
- The thermal coupling coefficient is related to bulk modulus and expansion.
- The equation of motion connects stress gradients with displacement.

Transition to next slide:

I will now define the mathematical scope used in the thesis.

Slide 13. Mathematical Scope and Primary Fields

Purpose:

Make the simplifications and primary fields explicit and defensible.

What to say:

The model is formulated as a two-dimensional continuum problem. The domain Ω is the model region, position is measured in metres, time is measured in seconds, and the displacement components are measured in metres. Small deformation and linear isotropic thermoelasticity are used as first-order approximations. The table also lists the primary fields: temperature, temperature perturbation, displacement, strain, stress, and direction variables. The model does not explicitly resolve pores, fractures, fluids, plasticity, damage, or full anisotropic constitutive behavior.

Key points:

- The thesis uses a two-dimensional continuum formulation.
- Small deformation and linear isotropic thermoelasticity are assumed.
- Complex geological effects are acknowledged as limitations.

Transition to next slide:

With the assumptions defined, I will explain the directional prediction task.

Slide 14. Directional Prediction Task

Purpose:

Define the source–probe query and what the prototype estimates.

What to say:

Directional prediction is built around a source point and a probe point. The vector from the source to the probe defines the direction of interest. Here x_s and x_p are positions in metres, d is the direction vector, L is distance, $\hat{d}$ is a dimensionless unit direction, and ϕ is the azimuth angle in radians. The mapping $\mathcal{F} : X \to \widehat{Y}$ means that the model maps input features to estimated outputs. We are not predicting every possible field value in a complete geological volume; we estimate directional characteristics for a specified query.

Key points:

- A source and a probe define the directional query.
- Distance and azimuth are explicit geometric quantities.
- The predicted response is interpreted along that direction.

Transition to next slide:

To train and compare the models, the work uses a controlled synthetic dataset.

Slide 15. COMSOL-Based Synthetic Dataset

Purpose:

Explain where the controlled training and comparison data come from.

What to say:

The dataset is derived from a controlled COMSOL-based thermoelastic calculation. The setup is a two-dimensional plane-strain rock domain with a heated internal source, used to generate time-dependent synthetic data. I will not focus on the initial conditions in the speech. The initial and boundary assumptions are used only to define a consistent calculation setting and are not the main focus of the presentation. The purpose of the dataset is to provide a common reference for comparing surrogate models under the same conditions.

Key points:

- The dataset comes from a controlled COMSOL-based calculation.
- The model uses a two-dimensional plane-strain domain.
- The data provide a shared basis for comparison.

Transition to next slide:

After data generation, the exports must be represented in a model-ready form.

Slide 16. COMSOL 2D Model and Mesh

Purpose:

Show the numerical domain and explain why the mesh matters for the dataset.

What to say:

This slide shows the two-dimensional COMSOL model and the finite-element mesh. The experiment represents a simplified section of a geological medium with an internal heated source. The mesh is refined near the source because this is where the largest thermal, displacement, and stress gradients are expected. This mesh is important because it defines the spatial structure from which the reference fields are exported.

Key points:

- The practical simulation is a simplified 2D domain.
- Mesh refinement is concentrated near the thermal source.
- The mesh provides the spatial basis for exported COMSOL fields.

Transition to next slide:
The next slide shows the temperature field generated by this setup.

Slide 17. COMSOL Temperature Field

Purpose:
Visualize the thermal part of the synthetic thermoelastic dataset.

What to say:
Here we see the temperature field from the COMSOL setup. The heated source creates a localized high-temperature region, while the surrounding area stays much closer to the initial temperature. This temperature field is one of the main dynamic variables used for model training and validation. It also provides the thermal driver for the mechanical response in the coupled problem.

Key points:

- The heat source creates a localized temperature disturbance.
- Temperature is one of the primary dynamic fields.
- The thermal field drives the coupled mechanical response.

Transition to next slide:
The next slide shows how this heating appears in displacement and stress.

Slide 18. COMSOL Thermoelastic Response

Purpose:
Connect the temperature field with displacement and stress outputs.

What to say:
The two images show the mechanical response produced by local heating. Thermal expansion creates displacement, and the constrained response of the surrounding medium creates stress concentration near the heated source. This confirms the coupled thermoelastic behaviour in the synthetic simulation. These fields are not field measurements; they are controlled numerical reference data for comparing neural surrogate models.

Key points:

- Heating produces displacement and stress concentration near the source.
- These fields demonstrate the coupled thermoelastic mechanism.
- The data are synthetic COMSOL references for model comparison.

Transition to next slide:

After generating the fields, they must be exported and preprocessed for the models.

Slide 19. Data Representation and Preprocessing

Purpose:

Show how raw simulation exports are prepared for different model families.

What to say:

The COMSOL comma-separated value exports are aligned into a single node set. Dynamic features are assembled into a spatio-temporal tensor $\mathbf{S}$ with dimensions for time steps, spatial points, and feature channels. Static material and geometry features are stored separately. Before training, the feature channels are normalized using the mean and standard deviation. In the normalization formula, $\tilde{x}$ is the normalized feature, x is the original feature in its native unit, and ε is a small stabilizer.

Key points:

- Raw exports are aligned by nodes and time steps.
- Dynamic and static features are represented separately.
- Normalization makes model training and comparison more stable.

Transition to next slide:

Now I will describe the comparative prototype framework.

Slide 20. Comparative Prototype Framework

Purpose:

Explain that all models are evaluated under one shared workflow.

What to say:

The prototype uses one workflow for all model routes. The frontend, backend orchestrator, material catalog, and model services follow a shared input-output contract. This is important for fair comparison: the same source–probe geometry, the same material presets, and the same normalized response format are used for every route. The goal is not to show a production platform, but to create a consistent research environment for comparing AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media methods.

Key points:

- All model routes receive comparable inputs.
- Outputs are normalized into the same response structure.
- The framework supports fair model comparison.

Transition to next slide:

Next, I will introduce the four model families compared in the work.

Slide 21. Shared Model Comparison Contract

Purpose:

Explain the shared input-output contract before discussing individual models.

What to say:

All routes are treated as surrogate modelling approaches for the same directional prediction task. The input side includes material parameters, source–probe geometry, observation time, and processed field features. The output side includes estimated temperature, displacement components, travel time, and diagnostic quantities. The comparison uses the same material presets, geometry contract, response format, and analytical travel-time sanity check. This keeps the comparison fair and avoids presenting the models as complete numerical solvers.

Key points:

- The comparison is based on one shared prediction contract.
- Inputs and outputs are kept consistent across all routes.
- The models are surrogate approaches, not equivalent physical solvers.

Transition to next slide:

I will now discuss each model route separately, starting with the physics-informed baseline.

Slide 22. Physics-Informed Neural Network (PINN)

Purpose:

Introduce the PINN route as the explicitly physics-informed baseline.

What to say:

The first model route is the Physics-Informed Neural Network. In the comparison it is the model that is closest to the thermoelastic formulation, because its training objective can include residuals connected with the governing equations. The input consists of coordinates, time, and material parameters such as Young's modulus, Poisson's ratio, density, thermal expansion, conductivity, and heat capacity. The output is the estimated temperature and in-plane displacement. The important point is balanced: PINN is physics-informed, but it is still a neural surrogate inside a simplified two-dimensional prototype, not an exact numerical solver.

Key points:

- PINN is the explicitly physics-informed route in the comparison.
- It maps coordinates, time, and material parameters to temperature and displacement.
- It is a baseline surrogate, not a complete equation solver.

Transition to next slide:

The next slides unpack how this PINN route is built and interpreted.

Slide 23. PINN: Physics-Informed Baseline

Purpose:

Clarify why PINN is treated differently from the other surrogate models.

What to say:

This slide states the main distinction between PINN and the other routes. PINN is called physics-informed because thermoelastic residual terms are part of the training loss. It is trained not only to match data, but also to reduce violations of selected physical relations. This does not make it a full COMSOL replacement. It means that, among the compared neural routes, PINN has the strongest direct link to the equations introduced in the thesis.

Key points:

- PINN combines supervised learning with physical residual terms.
- It is the reference physics-informed baseline for the comparison.
- The interpretation remains limited by the simplified 2D setup.

Transition to next slide:

Now I will show the input–output mapping used by the PINN.

Slide 24. PINN Input–Output Mapping

Purpose:

Explain what the network receives and what it predicts.

What to say:

The mapping is pointwise. For a given coordinate, observation time, and material parameters, the network predicts temperature and the two in-plane displacement components. Stress and strain are not primary outputs on this slide. They can be recovered from spatial derivatives of displacement and temperature-dependent constitutive relations. This is useful because the model can work with measurable field quantities while the physics residuals use derivatives internally.

Key points:

- Inputs: position, time, and effective material parameters.
- Outputs: temperature and in-plane displacement components.
- Stress and strain are derived diagnostics, not direct API outputs.

Transition to next slide:

The next slide shows the neural network structure behind this mapping.

Slide 25. PINN Network Structure

Purpose:

Describe the network architecture at a high level without overloading the defence.

What to say:

The PINN is represented as a multilayer perceptron. The input vector is passed through hidden layers with smooth activation functions. Smooth activations are important because the residual terms require spatial and temporal derivatives of the network output. Automatic differentiation is used to compute those derivatives. The details of every layer are less important than the reason for this structure: it supports differentiable field prediction.

Key points:

- The PINN is a differentiable multilayer perceptron.
- Smooth activations help compute PDE-related derivatives.
- Automatic differentiation provides derivatives for residuals.

Transition to next slide:

With the network defined, I will explain the composite loss.

Slide 26. PINN Composite Loss

Purpose:

Show how data fitting and physical consistency are combined.

What to say:

The composite loss contains data terms and physics terms. The supervised part compares predicted fields with reference data. The velocity consistency term supports dynamic interpretation. The elastic and thermal residuals encourage the network to respect the thermoelastic formulation. This is the main reason PINN is not grouped with purely supervised baselines. At the same time, the weights of the loss terms and the available data influence the final behaviour, so the output must still be validated.

Key points:

- The loss combines supervised, velocity, elastic, and thermal terms.
- Physics terms encourage consistency with thermoelastic relations.
- Loss balancing and data coverage remain important limitations.

Transition to next slide:

The next slide states the residuals more explicitly.

Slide 27. PINN Thermoelastic Residuals

Purpose:

Connect the PINN loss to the governing thermoelastic relations.

What to say:

This slide is the strongest connection between the implementation and the mathematical chapter. The residuals represent momentum balance, the thermoelastic stress relation, and the coupled heat equation. The coupling coefficient connects temperature change with volumetric strain and stress. In the defence I would not derive the equations again; I would simply emphasize that these terms are evaluated through automatic differentiation and added to the training objective.

Key points:

- The residuals use momentum balance and coupled heat behaviour.

- The coupling term links temperature, strain, and stress.
- This explicit residual use distinguishes PINN from the other routes.

Transition to next slide:

The last PINN detail slide explains its data use and final role.

Slide 28. PINN Data Use and Resulting Role

Purpose:

Summarize how PINN uses the dataset and why it is the main physical baseline.

What to say:

PINN uses the same COMSOL-derived two-dimensional data as the other models: coordinates, time, material parameters, and reference temperature and displacement. In addition, it evaluates residuals at collocation points. Therefore, it serves as the physically grounded reference route in the comparison. Its results are still interpreted within the training-data range and within the simplified two-dimensional assumptions.

Key points:

- PINN shares the COMSOL-derived dataset with the other routes.
- It also uses collocation points for residual evaluation.
- It is the strongest physics-informed baseline, not a full simulator.

Transition to next slide:

Next I will move to the Fourier Neural Operator route.

Slide 29. Fourier Neural Operator (FNO)

Purpose:

Introduce FNO as the structured-grid neural operator route.

What to say:

The Fourier Neural Operator is included as a neural-operator surrogate for structured two-dimensional fields. Its purpose is to learn a mapping between input fields and output response fields. This is different from PINN: the current FNO route uses physically meaningful fields, but it does not enforce the thermoelastic PDE residuals in the same explicit way. The current prototype also has an output-scale limitation, so FNO should be treated carefully as a diagnostic branch.

Key points:

- FNO is a field-to-field neural operator route.
- It works with physical fields but is not explicitly physics-informed like PINN.
- The current implementation requires recalibration.

Transition to next slide:

The next slides explain how the FNO route represents fields.

Slide 30. FNO: Field-to-Field Prediction Idea

Purpose:

Explain the operator-learning view behind FNO.

What to say:

The key idea is that the output of a thermoelastic simulation is a spatial field, not just a scalar number. FNO learns an operator from an input field space to an output field space. This makes it attractive for grid-based temperature and displacement prediction. In this thesis it is used as a neural operator baseline, meaning it learns from data and field structure rather than from explicit thermoelastic residual enforcement.

Key points:

- The model learns a mapping between function spaces.
- Inputs and outputs preserve spatial grid structure.
- It is a neural operator baseline for comparison.

Transition to next slide:

Now I will show the tensor format used in the 2D setup.

Slide 31. FNO Tensor Format for the 2D Setup

Purpose:

Document the practical two-dimensional tensor representation.

What to say:

The FNO receives tensors with batch, channel, and two spatial dimensions. This matches the practical two-dimensional setup of the thesis. The output tensor has the same spatial grid and a selected number of output channels. In the simplified thermoelastic setup, those channels can represent temperature and in-plane displacement components. This is another place where the difference between theory and implementation is visible: the theory is general, but this implementation is two-dimensional.

Key points:

- Input and output tensors are two-dimensional grids.
- Channels represent physical quantities such as temperature and displacement.
- The practical setup remains 2D.

Transition to next slide:

The next slide gives the main architecture of the FNO route.

Slide 32. FNO Architecture Used in the Prototype

Purpose:

Summarize the FNO processing chain.

What to say:

The FNO architecture has four main stages: lifting, Fourier blocks, projection, and output. The lifting layer maps input channels to a latent representation. Fourier blocks process the latent field using spectral and local branches. The projection layer maps the latent representation back to output channels. The implementation parameters on the slide define the retained Fourier modes, number of Fourier layers, and channel capacity.

Key points:

- FNO processes fields through lifting, Fourier blocks, and projection.
- Fourier modes control spectral capacity and cost.
- The architecture is suitable for regular 2D grids.

Transition to next slide:

The next slide explains the spectral block itself.

Slide 33. FNO Spectral Block

Purpose:

Explain how the spectral and local branches work together.

What to say:

A Fourier block combines a spectral branch with a local convolution branch. The spectral branch transforms the latent field into Fourier space, applies learned complex-valued weights to selected coefficients, and transforms the result back. This helps capture broader field patterns. The local branch keeps neighbouring spatial information. Together, they allow the model to learn both global and local structure in the field.

Key points:

- The spectral branch works in Fourier space.
- The local convolution branch handles neighbouring patterns.
- The block is data-driven, not a PDE residual calculation.

Transition to next slide:

Now I will explain mode truncation and coordinate encoding.

Slide 34. Mode Truncation and Coordinate Encoding

Purpose:

Clarify two practical FNO design choices.

What to say:

Mode truncation means that only selected low-frequency Fourier coefficients are transformed with learned weights. This reduces computational cost and controls model capacity. Coordinate encoding adds normalized spatial coordinates as extra channels, so the model can distinguish source, boundary, and interior regions. These are engineering choices that help the neural operator learn spatially organized behaviour.

Key points:

- Retained modes control spectral resolution and cost.
- Coordinate channels give the model spatial awareness.
- These choices support learning but do not enforce physics directly.

Transition to next slide:

The last FNO slide states the training objective and caveat.

Slide 35. Training, Evaluation, and Current FNO Caveat

Purpose:

Make the FNO interpretation honest before the result slides.

What to say:

The FNO objective is supervised: it combines mean squared error with a relative error term. Channel normalization is computed on the training set and then reused for validation and inference. In the current prototype, the FNO branch still has an output-scale calibration issue. Therefore, it is useful for showing the neural-operator route, but its physical predictions should not be overclaimed until recalibration is completed.

Key points:

- The current FNO loss is supervised MSE plus relative error.
- It uses normalization of physical channels.
- Its output scale needs recalibration before strong physical claims.

Transition to next slide:

Next I will describe the MeshGraphNet route.

Slide 36. MeshGraphNet

Purpose:

Introduce MeshGraphNet as the mesh-aware surrogate route.

What to say:

MeshGraphNet is the graph-based route. It is included because COMSOL data are naturally mesh-based: nodes and elements form local neighbourhoods. The model can use mesh nodes, connectivity, material features, and query context to estimate node-level or probe-level responses. In this prototype it is best described as a mesh-aware surrogate candidate. Its current limitation is temporal: it behaves mainly as a fixed-horizon predictor.

Key points:

- MeshGraphNet represents the computational domain as a graph.
- It can preserve mesh connectivity more naturally than a grid model.
- The current route needs stronger time-conditioning.

Transition to next slide:

The next slides unpack the graph representation.

Slide 37. MeshGraphNet: Why a Graph Model

Purpose:

Explain why a graph model is relevant for COMSOL-derived data.

What to say:

Finite-element data do not always lie on a regular image grid. A graph representation preserves node connectivity, nonuniform distances, and local neighbourhood structure. This matters because heat transfer, stress redistribution, and elastic response depend on local geometry. MeshGraphNet is therefore a natural candidate for mesh-aware surrogate modelling, although it still learns from data rather than explicitly solving the PDE system.

Key points:

- COMSOL mesh data are naturally represented as nodes and edges.
- Graphs preserve local neighbourhood geometry.
- The route is supervised and mesh-aware, not a full physics solver.

Transition to next slide:

Now I will show the edge features.

Slide 38. Mesh and Edge Features

Purpose:

Show how geometry enters the graph model.

What to say:

The edge features describe the relative vector and distance between neighbouring mesh nodes. This gives the model information about both direction and separation length. These geometric features are important because the influence between two neighbouring nodes depends not only on their state values, but also on their spatial relation. In simple terms, the graph edges carry local geometry into the neural model.

Key points:

- Edge features include relative position and distance.
- Geometry helps represent local physical neighbourhoods.
- The graph model avoids forcing all data onto a regular grid.

Transition to next slide:

The next slide shows node features and the prediction target.

Slide 39. Node Features and Prediction Target

Purpose:

Clarify what each graph node contains and what the model predicts.

What to say:

Each node feature vector can include coordinates, material parameters, scenario conditions, and current dynamic fields. The model then predicts a future field matrix over all nodes. This makes MeshGraphNet suitable for state-transition style prediction. The important distinction is that the physical quantities are present as features and targets, while the governing equations are not enforced as explicit residuals in the same way as PINN.

Key points:

- Nodes contain geometry, material information, and current state variables.
- The target is the future field over graph nodes.
- Physics enters through features and data, not explicit residual enforcement.

Transition to next slide:

Now I will explain absolute and delta prediction modes.

Slide 40. Delta Prediction Mode

Purpose:

Explain why predicting state changes can be useful.

What to say:

MeshGraphNet can predict either the absolute next state or the change from the current state. Delta prediction is often useful because changes between neighbouring time steps are smoother and smaller than full field values. However, there is a rollout caveat: in autoregressive prediction, each predicted state becomes the next input, so small one-step errors can accumulate over longer horizons.

Key points:

- Absolute mode predicts the next state directly.
- Delta mode predicts the change from the current state.
- Autoregressive rollout can accumulate errors.

Transition to next slide:

The next slide gives the architecture of the graph network.

Slide 41. Encoder–Processor–Decoder Structure

Purpose:

Describe the MeshGraphNet architecture without excessive implementation detail.

What to say:

The architecture has three parts. The encoder maps raw node and edge features into latent vectors. The processor applies repeated message passing over the graph connectivity. The decoder converts the final latent node states into predicted fields. This encoder–processor–decoder design is common in graph neural simulators because it separates feature embedding, interaction modelling, and output prediction.

Key points:

- Encoder: embeds raw node and edge features.
- Processor: performs message passing over the graph.
- Decoder: maps latent states to predicted fields.

Transition to next slide:

The last MeshGraphNet slide explains message passing and limits.

Slide 42. Message Passing, Training, and Limits

Purpose:

Connect graph message passing with training and interpretation.

What to say:

Message passing updates edge and node representations by exchanging information along graph edges. This can be interpreted as a learned local interaction mechanism. The training objective is supervised and compares predicted node fields with reference fields, possibly with channel weights. Therefore, MeshGraphNet is a supervised COMSOL surrogate: it preserves mesh topology and supports material conditioning, but the physical equations are learned implicitly and must be checked through validation plots and metrics.

Key points:

- Message passing learns local interactions on the mesh graph.
- Training is supervised against reference fields.
- Physical reliability depends on validation, not residual enforcement.

Transition to next slide:

The fourth model route is the Transformer-style surrogate.

Slide 43. Transformer-Style Surrogate Model

Purpose:

Introduce the attention-based surrogate route.

What to say:

The Transformer-style model is an attention-based baseline surrogate for fast query prediction. Its input is a set of tokens describing geometry, material parameters, state variables, and requested observation time. Its output is a fast estimate of temperature, displacement, travel time, and diagnostics. It is included because it is flexible and low-latency. Its limitation is that it has fewer explicit physics priors than the PINN route, so results should be interpreted cautiously.

Key points:

- The Transformer route is a supervised attention-based surrogate.

- It uses tokens for geometry, material data, and state variables.
- Its physics constraints are weaker than in the PINN route.

Transition to next slide:

The next slides explain how tokenisation and attention are used.

Slide 44. Transformer: Attention-Based Surrogate

Purpose:

Explain why attention is useful for query-style prediction.

What to say:

This route treats sampled points as tokens. Attention allows each query point to relate to other points in the domain, including distant points. This makes the model flexible for query-style prediction at requested locations. In this thesis it is not presented as a physics-informed solver. It is a supervised operator-style baseline whose physical meaning comes from its inputs, outputs, and training data.

Key points:

- Points are represented as tokens.
- Attention can connect distant regions of the domain.
- The route is data-driven and should be validated carefully.

Transition to next slide:

Now I will describe the tokens used by the model.

Slide 45. Transformer Tokenisation

Purpose:

Define the input and query tokens in the Transformer route.

What to say:

Input tokens describe known sample points. They include coordinates, initial temperature and displacement, velocity-like components, and material parameters. Query tokens describe where the model should predict the response. This token structure makes the model flexible: the same network can combine physical state information with requested prediction locations.

Key points:

- Input tokens contain state, geometry, and material information.
- Query tokens mark requested prediction locations.
- Tokenisation converts the physical field problem into an attention problem.

Transition to next slide:

The next slide gives the attention operation.

Slide 46. Scaled Dot-Product Attention

Purpose:

Explain the attention formula at an intuitive level.

What to say:

Scaled dot-product attention compares queries and keys to compute weights, then uses those weights to combine value vectors. Intuitively, this lets a requested point decide which input points are most relevant for its prediction. The score matrix connects every query with every input token. This is useful for long-range dependencies, but it is still a learned statistical mechanism rather than an explicit physical equation.

Key points:

- Queries, keys, and values define weighted information exchange.
- Attention supports long-range interactions.
- It does not by itself enforce thermoelastic conservation laws.

Transition to next slide:

Now I will show the encoder and decoder structure.

Slide 47. Transformer Encoder and Decoder

Purpose:

Summarize the architecture used for query prediction.

What to say:

The encoder processes input tokens using self-attention and feed-forward layers. The decoder uses query tokens and cross-attention to attend to the encoded input representation. This means the model first builds a representation of the known thermoelastic state and then predicts the requested response at query points. This is why it can be described as a supervised sequence or operator-style baseline.

Key points:

- Encoder: processes known input tokens.
- Decoder: uses query tokens and cross-attention.
- The architecture supports flexible query-point prediction.

Transition to next slide:

The next slide shows the output head.

Slide 48. Transformer Output Head

Purpose:

Clarify the predicted fields and possible derived diagnostics.

What to say:

The output head predicts temperature and two in-plane displacement components at query points. Stress and strain are not direct outputs here. They can be derived later from displacement gradients if sufficient spatial information is available, or they could be added as future output channels. This keeps the current route aligned with the shared output contract without overstating its physical enforcement.

Key points:

- Outputs are temperature and in-plane displacement at query points.
- Stress and strain are derived or future diagnostics.
- The model is fast but less physically constrained than PINN.

Transition to next slide:

The final Transformer slide describes the training objective.

Slide 49. Transformer Training Objective

Purpose:

State how the Transformer route is trained and how to interpret it.

What to say:

The slide shows a hybrid supervised objective with reconstruction, velocity, and thermal regularisation terms. The practical interpretation remains cautious: the Transformer route is mainly a supervised surrogate. Token subsampling helps control attention cost and encourages operation on different spatial samplings. Its predictions should be checked against COMSOL-derived references and should not be described as solving the full thermoelastic PDE system.

Key points:

- Training is mainly supervised with additional regularisation terms.
- Token subsampling controls computational cost.
- The route is not physics-informed in the strict PINN sense.

Transition to next slide:

Now I will move from model descriptions to the calculation setup and results.

Slide 50. Calculation Status

Purpose:

Summarize the evaluation setup and make caveats transparent before showing plots.

What to say:

The calculation setup uses forty scenarios per material and four materials: granite, sandstone, limestone, and basalt. With four model routes, this gives six hundred forty model responses under the same contract. Every plotted response is an active model output, not a mock fallback. The operating envelope is still limited to the two-dimensional plane-strain setup. The Fourier Neural Operator is included, but its output scale is currently miscalibrated, so it should be interpreted as a diagnostic branch rather than a reliable final route.

Key points:

- The comparison covers $40 \times 4 \times 4 = 640$ model responses.
- All plotted responses are active outputs.
- FNO is included, but its current calibration is limited.

Transition to next slide:

The first result concerns predicted travel times.

Slide 51. Travel-Time Predictions

Purpose:

Explain what the travel-time plot shows and what conclusion should be drawn.

What to say:

This plot compares predicted travel time by model and material. The main observation is that the differences between model families are larger than the difference between basalt and sandstone in the current prototype. PINN and Transformer produce much smaller travel-time estimates, while FNO and MeshGraphNet produce larger values. This does not mean one model is universally correct. It shows that, under the current setup, model formulation strongly influences directional travel-time estimates. Therefore, model comparison is necessary before drawing physical conclusions.

Key points:

- The plot compares travel-time estimates across models and materials.
- Model-family differences dominate the material difference here.
- The result motivates careful interpretation and validation.

Transition to next slide:

To judge accuracy, the next slide compares the estimates with an analytical reference.

Slide 52. Accuracy Against Analytical Reference

Purpose:

Present the key quantitative result and its cautious interpretation.

What to say:

Here the predicted travel times are compared with a simple analytical reference, $t_{\text{gt}} = r/V_P$. In this formula, t_{gt} is reference travel time in seconds, r is source–probe distance in metres, and V_P is compressional wave velocity in metres per second. This reference is not a full thermoelastic validation; it is a useful sanity check. In this comparison, PINN has the lowest median relative error and remains within the ten percent band across the balanced sweep.

Key points:

- The reference is a sanity check, not complete validation.
- PINN shows the best agreement with the analytical reference.
- Other routes need further calibration or time-conditioning.

Transition to next slide:

Accuracy is only one side of the comparison; inference time is also important.

Slide 53. Inference Latency and Time Behavior

Purpose:

Compare practical response time and explain the fixed-horizon issue.

What to say:

This plot shows inference latency by model. All routes return predictions in milliseconds on the development machine, so the prototype is suitable for interactive comparison. The Transformer route is the fastest, followed by MeshGraphNet, FNO, and PINN. However, latency is not the only criterion. PINN is also the only route that changes smoothly with requested observation time. The other routes currently behave more like fixed-horizon predictors. This means they may be fast, but their temporal behavior needs additional development for time-dependent directional studies.

Key points:

- All routes are fast enough for interactive prototype use.
- Transformer has the lowest median latency.
- PINN currently has the most meaningful time-conditioned behavior.

Transition to next slide:

Before concluding, I will state the main threats to validity.

Slide 54. Threats to Validity

Purpose:

Show scientific caution and define the limits of the results.

What to say:

There are several threats to validity. First, the training data are limited and come from a pilot COMSOL-derived configuration, not from a complete calibrated geological database. Second, the model uses two-dimensional plane-strain assumptions and effective material parameters. Third, the current Fourier Neural Operator route has a normalization issue and is not yet physically reliable. Fourth, the analytical reference based on r/V_P is only a sanity check, not a full validation of thermoelastic behavior. These limitations are important because the results should be interpreted within the simplified assumptions of the study.

Key points:

- The dataset is limited and synthetic.
- The physical formulation is simplified to two-dimensional effective media.
- The analytical reference is useful but limited.

Transition to next slide:

With these limitations in mind, I can summarize the conclusions.

Slide 55. Conclusions

Purpose:

State the main scientific outcomes of the work.

What to say:

The thesis develops a physically grounded comparative framework for AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media. Within the tested assumptions, PINN is the strongest physics-informed baseline and shows the best agreement with the analytical travel-time reference. Transformer-style prediction is the fastest route and is useful for rapid query-based inference. MeshGraphNet remains a useful mesh-aware surrogate candidate, but needs further time-conditioning. FNO should be treated as a diagnostic branch until its output normalization is corrected. Overall, the work demonstrates a careful prototype for comparing neural surrogate approaches rather than a final geophysical simulator.

Key points:

- The main contribution is a comparative AI-assisted framework.
- PINN performs best against the travel-time reference.
- Other models remain useful candidates with specific limitations.

Transition to next slide:

Finally, I will outline the most important next steps.

Slide 56. Future Work

Purpose:

Show how the prototype can be developed after the diploma.

What to say:

The first direction is to retrain and recalibrate the two-dimensional FNO route, because its current output scale is not reliable. The second direction is to add stronger time-conditioned rollout behavior to MeshGraphNet. The third direction is to extend the material catalogue with more complete thermoelastic parameters and validation data. It would also be useful to compare the prototype against additional analytical and numerical baselines. The final message is that the prototype estimates directional characteristics of thermoelastic wave propagation within simplified assumptions and provides a basis for further research.

Key points:

- Recalibrate FNO and improve time behavior in MeshGraphNet.
- Extend material data and validation sources.
- Add stronger analytical or numerical baselines.

Transition to next slide:

This completes the presentation, and I am ready for questions.

Slide 57. Thank You

Purpose:

Close the presentation and invite questions.

What to say:

Thank you for your attention. I will be happy to answer your questions about the physical assumptions, the model comparison, the dataset, or the interpretation of the results.

Key points:

- Close politely.
- Signal readiness to discuss assumptions and limitations.
- Keep answers consistent with the research prototype framing.

Transition to next slide:

Questions from the committee.

Possible questions from the committee

1. Why did you choose directional prediction instead of full-field simulation?

Because the thesis focuses on a narrower and more defensible research task. Full-field thermoelastic simulation is computationally and physically more complex. Directional prediction allows us to compare source–probe responses, travel time, and model behavior under controlled assumptions.

2. Is the prototype validated on real field data?

No. The current work uses COMSOL-derived synthetic data and analytical sanity checks. The results should be interpreted within simplified modelling assumptions. Field data would be an important next step, but it is outside the current scope.

3. Why is PINN called a physics-informed baseline?

PINN is called a physics-informed baseline because its training objective includes residual terms connected with thermoelastic relations. This means that physical structure is included in the learning objective, but the result is still interpreted as a baseline model inside the simplified prototype.

4. Why does FNO perform poorly in the current results?

The current two-dimensional FNO route has a calibration and output-scale issue. Therefore, it is kept in the comparison as a diagnostic branch. Its architecture is still relevant, but this implementation needs retraining and normalization correction.

5. What is the role of the analytical reference $t = r/V_P$?

It is a simple travel-time sanity check based on distance and P-wave velocity. It is not a complete thermoelastic validation, but it helps identify whether predicted travel times are physically plausible in a basic isotropic approximation.

6. What are the main limitations of the work?

The main limitations are the simplified two-dimensional setting, effective isotropic material parameters, limited synthetic data, fixed-horizon behavior in some surrogate routes, and the absence of direct calibration against real geological measurements.

7. What is the main contribution of the diploma work?

The main contribution is a physically grounded comparative framework for AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media. It combines thermoelastic theory, geological material parameters, a controlled dataset, and several neural surrogate routes under one consistent comparison workflow.

Short version of the speech

Good morning. The topic of our diploma work is *AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media*. The work presents a research prototype for AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media. It is not a site-calibrated simulator. The goal is to estimate selected directional characteristics of thermoelastic wave propagation under simplified assumptions and to compare several neural surrogate approaches.

The physical motivation is that temperature changes in rocks can create thermal expansion, stress, displacement, and wave-like mechanical response. This is relevant for geothermal systems, underground engineering, and geophysical interpretation. A full coupled numerical simulation can be expensive when many materials and source–probe directions must be compared. Therefore, the thesis formulates a narrower prediction task.

The model uses effective geological material parameters such as density, elastic moduli, thermal conductivity, heat capacity, thermal expansion, and P-wave and S-wave velocities. These parameters describe how different rock types respond mechanically and thermally. The simplified formulation is two-dimensional, uses small deformation and linear isotropic thermoelasticity, and does not explicitly resolve pores, fluids, fractures, plasticity, or full anisotropy.

Directional prediction is defined by a source point and a probe point. Their difference gives the direction, distance, and azimuth. The prototype estimates temperature, displacement, travel time, and directional response for that query. The data are based on a controlled COMSOL-derived two-dimensional thermoelastic setup, exported and normalized for several model families.

The comparison includes the Physics-Informed Neural Network, Fourier Neural Operator, MeshGraphNet, and a Transformer-style route. PINN is used as the physics-informed baseline because its objective includes thermoelastic residual terms. FNO, MeshGraphNet, and Transformer-style models are treated as surrogate routes for comparison, not as physically equivalent numerical solvers.

The calculations use forty scenarios per material, four materials, and four model routes, giving six hundred forty model responses. In the travel-time comparison, model-family differences are larger than the material difference in the current prototype. Against the simple analytical reference $t = r/V_P$, PINN shows the best agreement and remains within the ten percent band in the balanced sweep. Transformer-style prediction is the fastest route, while MeshGraphNet and FNO need additional development for time behavior and calibration.

The main limitations are the limited synthetic dataset, the simplified two-dimensional assumptions, the current FNO normalization issue, and the fact that the analytical reference is only a sanity check. The conclusion is that the work provides a physically grounded comparative framework for AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media within simplified assumptions. Future work should recalibrate FNO, improve time-conditioned MeshGraphNet behavior, extend material data, and compare against additional analytical or numerical baselines.